Malik Muhammad Saad

Physical Embodied AI Researcher | **5G/6G Wireless Networks**

Ph.D. in Computer Science & Engineering, Kyungpook National University, South Korea

maliksaad84@gmail.com | **+82 10 6604 9603**

Personal Webpage | ORCID | Google Scholar | ORCID | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Physical Embodied AI Researcher currently affiliated with the **CORE-AI Lab (Physical Embodied AI)**, conducting research on **semantically grounded world models**, **Vision-Language Models (VLMs)**, **Large Language Models (LLMs)**,**Vision-Language-Action (VLA)** systems, **reinforcement learning**, and **embodied intelligence** for autonomous vehicles and dexterous robotic manipulation. Experienced in developing **multimodal perception**, **structured reasoning**, **persistent semantic memory**, and **reinforcement learning–based post-training** techniques for safety-critical AI systems.

Strong background in **AI-driven mobility** and communication systems spanning **5G/6G**, **NR-V2X**, **distributed AI**, **intelligent transportation systems (ITS)**, and **Connected and Cooperative Automated Vehicles (C-CAV)**. Proficient in **software development**, **large-scale simulation**, and practical implementation of wireless communication systems using **Software-Defined Radios (SDRs)**, **GNU Radio**, **Python**, **MATLAB**, **MuJoCo**, and **ns-3**. Experienced in **Deep Reinforcement Learning (TD3, DDPG, PPO, SAC)**, **multimodal learning**, **representation learning**, and **foundation model development**, with expertise in translating research into deployable prototypes and experimental platforms.

Actively engaged in **3GPP-aligned R&D** and **5GAA** collaborative research, with contributions to **intelligent resource allocation**, **sidelink communications**, **vehicular networking**, and **AI-enabled communication architectures**. Published research in leading **IEEE journals**, **IEEE magazines**, and **international conferences** across artificial intelligence, wireless communications, autonomous systems, and embodied AI.

TECHNICAL SKILLS

- **Software Defined Radios (SDRs):** USRPs, AntSDR, HackRF, RTL-SDR, BladeRF, Pluto SDR
- **Embedded Systems & Boards:** NVIDIA Jetson Orin Nano, Raspberry Pi, PIC Microcontrollers, LoRaWAN Modules
- **Mobile AI Platforms:** Google Pixel (Tensor SoC), Snapdragon-based devices, heterogeneous computing (CPU, GPU, NPU)
- **AI/ML:** Reinforcement Learning, Deep Reinforcement Learning (DRL), LSTM, CNN, Supervised Learning, Transfer Learning
- **Frameworks:** TensorFlow, PyTorch, Scikit-learn, Matplotlib
- **Simulation Tools:** ns-3, MATLAB, SUMO, GNU Radio, CARLA, MuJuCo, Simulink, Wireshark
- **OS/RTOS:** Linux, QNX, VMware
- **Contribution to Intellectual Property Rights (IPR)**

PROGRAMMING SKILLS

- C++
- C
- Python
- Arduino
- RESTful API, MQTT, SQL
- Linux System Programming

DOMAIN EXPERTISE

- C-V2X / NR-V2X, Resource Allocation, Cooperative-Connected Automated Vehicles (C-CAVs), On-device AI, Edge Intelligence, Sensor Fusion, Semantics V2X Communications, NTN, Network Slicing, O-RAN /AI-RAN.

- **Standards:** 3GPP, IEEE, ETSI, ITS, ASPICE, and ISO/IEC 4200.

FULL-TIME EMPLOYMENT EXPERIENCE (8+ YEARS)

1. Physical Embodied AI Researcher (InnoCORE Research Fellow, Ministry of Science and ICT, Korea)

Organization: CORE-AI Lab, DGIST, Korea
Duration: February 2026 – Present

- Conducting research on **on-device AI systems** for **Physical Embodied AI**, focusing on efficient execution of LLMs under real-world hardware constraints.
- Researching **Generative AI** and **multimodal foundation models** for autonomous driving, robotics, and embodied intelligence.
- Developing **semantically grounded 3D world models** by integrating occupancy prediction, vision-language representations, and temporal memory for autonomous perception.
- Designing **persistent semantic memory architectures** for open-vocabulary 3D scene understanding and long-horizon environmental reasoning.
- Investigating **latent reasoning architectures** that replace language-token reasoning with latent-space inference for efficient multimodal reasoning.
- Developing **structured reasoning mechanisms** to mitigate hallucinations in Vision-Language Models (VLMs) for safety-critical autonomous systems.
- Applying **reinforcement learning (RL)** and **post-training alignment** techniques to improve reasoning robustness and decision-making capabilities of foundation models.
- Building **Vision-Language-Action (VLA)** models for dexterous robotic manipulation using multimodal sensory inputs, including vision, language, proprioception, and tactile feedback.
- Designing large-scale simulation pipelines in **MuJoCo** for synthetic dataset generation supporting embodied AI and robotic foundation model training.
- Developing **closed-loop tactile-aware manipulation policies** for dexterous grasping through multimodal sensor fusion and feedback-driven control.
- Conducting research on **multimodal learning**, **representation learning**, autonomous systems, embodied AI, and foundation models for next-generation intelligent agents.

2. Postdoctoral Researcher at Kyungpook National University, Korea

Organization: Kyungpook National University, Korea
Duration: 01 March 2025 – 31 January 2026

- Leading research on AI-driven distributed resource allocation protocols for 5G/6G-based NR-V2X communication systems to complement 3GPP standards.
- Developed Agentic AI–enabled V2X frameworks that extract semantics from the perception layer and apply reasoning mechanisms for context-aware transmission to enable C-CAV.
- Designed simulation and testing pipelines using GNU Radio, ns-3, and SUMO, emphasizing latency-critical vehicular safety applications.
- Conducted investigations on network slicing and intelligent resource management to enable scalable, service-oriented vehicular communication frameworks.
- Worked with Qualcomm V2X SDK-aligned frameworks, mapping MAC behavior to Snapdragon automotive SoCs and applying real-time constraints in RTOS environments (QNX, Linux).
- Implemented application-level messaging protocols (CAM, DENM, BSM) and conducted protocol stack validation for safety-critical communication.
- Coordinated joint research with the University of Glasgow and 5GAA, aligning findings with 3GPP Release 17/18 standardization.
- Published research in high-impact IEEE journals, conferences, and magazines, addressing challenges in intelligent vehicular networking, edge computing, and AI-based communication systems.
- Actively participated in 3GPP activities, contributing to technical discussions and reports on sidelink scheduling, NR-V2X enhancements, and 6G architecture development.

3. Adjunct Faculty at Dongseo University, Busan, Korea

Organization: Dongseo University, Busan, Korea

Duration: 02 March 2025 – **Present**

- Delivered undergraduate-level courses in Operating Systems and Database Systems, covering memory management, process scheduling, SQL, relational modeling, and transaction processing (Spring Semester: March – June 2025).
- Delivering undergraduate-level courses on Computer Networks, Network Programming, and Programming Method (Fall Semester: September – December 2025).

4. Research Assistant at Wireless and Mobile Internet Laboratory (MONET)

Organization: Kyungpook National University, Korea
Duration: 01 Sep 2019 – 28 Feb 2025

- Conducted research in wireless networks specializing in vehicular communications (C-V2X and NR-V2X).
- Contributed to multiple implementation-driven projects focused on distributed scheduling, mobility modeling, and AI-based resource allocation.
- Investigated network slicing, Network Function Virtualization (NFV), and radio virtualization to enhance flexibility and efficiency in next-generation vehicular networks.
- Served in editorial and organizational roles for IEEE flagship conferences on behalf of the supervisor.
- Mentored Master's and Ph.D. students, providing guidance on research design, implementation, and documentation.

5. Research Assistant at Institute of Space Technology (IST), Pakistan

Organization: Institute of Space Technology, Pakistan
Duration: 02 July 2018 – 02 April 2019

- Worked on a funded research project titled "5G Cognitive Cells Deployment," supported by the Ministry of Science & Technology (ICT R&D Fund).
- Conducted experimental evaluation of 5G cognitive cell architectures using Software Defined Radios (SDRs), GNU Radio, and Linux-based platforms.
- Implemented and tested radio virtualization using OFDM and FBMC physical layer waveforms.
- Developed and validated a single radio interface capable of supporting multiple virtual radios through resource abstraction.
- Contributed to research on software-defined and virtualized RANs for enhanced spectrum utilization.

6. Teaching Research Associate (Faculty) at Institute of Space Technology, Islamabad, Pakistan

Organization: Institute of Space Technology, Pakistan
Duration: 02 April 2019 – 25 August 2019

- Taught Embedded Systems and Digital Logic Design courses with hands-on lab sessions.
- Delivered practical knowledge of microcontroller programming using C and Assembly.

7. Application Design RF Engineer at Micro Electronics International (Pvt) Ltd., Islamabad, Pakistan

Organization: Micro Electronics International (Pvt) Ltd., Islamabad
Duration: 02 Nov 2018 – 30 March 2019

- Developed SDR-based tactical communication prototypes for defense applications.
- Configured network routers and radio base stations for field operations.
- Participated in technical bidding and tender preparation, contributing to system design documentation, compliance specifications, and cost estimation.
- Managed technical reports and documentation, maintaining accurate records of project performance and validation results.

KEY PROJECTS & RESEARCH CONTRIBUTIONS

1. AI-Based Congestion Control in NR-V2X MAC Layer

Kyungpook National University & University of Glasgow (2023–2024)

- Designed and implemented a deep reinforcement learning-based resource allocation scheme for

congestion control in NR-V2X.
- Incorporated both system-level metrics (Channel Busy Ratio, Channel Occupancy Ratio) and application-level Age-of-Information (AoI).
- Achieved dynamic optimization of transmission rate, power control, and resource reservation intervals, improving network throughput and reliability.
- Work aligns with Qualcomm's V2X SDK, designed for Snapdragon SA2150P / SA8155P chipsets, with transferable scheduling logic targeting ARM-based SoCs and real-time execution environments such as QNX.

2. Cooperative Scheduling Framework for 5G NR-V2X
Kyungpook National University & University of Glasgow (2022–2023)

- Conducted performance evaluation of NR-V2X enhancements over legacy C-V2X using a custom NR-V2X simulator built in ns-3 (sub-6 GHz).
- Proposed practical improvements for decentralized resource selection and cooperative MAC layer scheduling under high-mobility scenarios.

3. SDR-Based Evaluation of Cognitive 5G Cells
Institute of Space Technology (IST), Pakistan (2018)

- Experimentally validated Software Defined Networking (SDN) and Network Functions Virtualization (NFV) for dynamic network slicing.
- Developed and deployed Virtual Radios (VRs) using FBMC and OFDM air interfaces to ensure low-latency and high-reliability communication.
- Conducted full-stack real-time SDR testing using GNU Radio and USRP platforms under Linux, integrating Python/C++ modules and flowgraphs for performance validation.
- Demonstrated flexibility of software-based 5G cognitive cells through real-time SDR testing.

4. mmWave Backhaul via High-Altitude Platforms (HAP-UAV)
Institute of Space Technology (IST), Pakistan (2018)

- Engineered backhaul communication links using mmWave (28/31 GHz) bands for remote regions in Pakistan.
- Demonstrated wireless delivery of internet services using UAV-based HAPs as airborne relays.
- Validated feasibility of HAP-based backhaul in rural and disaster-prone environments.

PROFESSIONAL & COLLABORATIVE SKILLS

- **3GPP Standardization Participation:** Active participant in **3GPP RAN Working Groups (RAN1 / RAN2 / RAN4)**, contributing to ongoing standardization efforts for **V2X communication protocols**, particularly focusing on **NR-V2X Mode 2** and **PC5 sidelink enhancements**.
- **Software Development:** Developed advanced **software modules** and end-to-end **simulation frameworks** for testing and validating next-generation V2X systems. Designed a **Semantic Communication Simulation Framework** for 6G-V2X using **Agentic AI**–based perception and reasoning layers to enable contextual and goal-oriented message transmission for C-CAVs. Additionally, implemented multiple **3GPP-compliant validation frameworks** for NR-V2X Mode 2, including conflict-free resource allocation, sidelink congestion control, SPS/RRI/RC evaluation, and **Age-of-Information–aware** scheduling mechanisms.
- **Technical Proposal Writing:** Experienced in drafting and contributing to competitive **R&D proposals** for international research grants, including **NRF Korea** and **EU Horizon** calls. Skilled in writing **project objectives**, **technical scopes**, and **milestone plans** aligned with real-world **industry impact**.
- **Conference Presentations:** Presented research at international seminars and **IEEE flagship conferences**, including **ICC**, **ICUFN**, **ICTA**, **ICAIIC**, and other **IEEE Transactions**-affiliated events. Proven ability to effectively communicate complex technical ideas to diverse academic and industrial audiences.
- **Global Collaboration:** Worked on **multi-national research projects** with collaborators from the **UK**, **South Korea**, and **Pakistan**, including partnerships with the **University of Glasgow** and

Coventry University.

- **Technical Peer Review:** Served as a **Reviewer** and **Technical Program Committee (TPC) Member** for reputed journals and conferences under **IEEE**, **ACM**, and **Elsevier**.

TRAINING WORKSHOPS

1. Intern at the National Institute of Electronics (Ministry of Information Technology)

Duration: 01 July 2016 – 29 July 2016

- Designed and implemented software for embedded systems and microcontroller-based devices.
- Involved in developing, coding, testing, and debugging low-level system software for real-time applications.

2. Intern at the Institute of Space Technology, Islamabad

Duration: 18 July 2017 – 18 August 2017

- Gained hands-on experience with RTL-SDR receivers and USRP-based communication setups.
- Explored multi-hop wireless communication and practical SDR experimentation for radio signal analysis.

EDITORIAL ACTIVITIES

Lead Guest Editor, IEEE Communications Standards Magazine

Special Issue Title: Toward 6G NTN-V2X: Architectures, Standardization Roadmap, and AI-Driven Technologies
Status: Officially accepted and published online in July 2025
Link: View Special Issue Online

- Led the proposal, coordination, and launch of a feature topic on Non-Terrestrial Networks (NTN) for V2X.
- Collaborated with global experts, including contributors from **5GAA** and other leading research institutes and industries.
- Managed the editorial process via ScholarOne, organized calls for papers, reviewer invitations, and ensured technical scope alignment with **3GPP standards**.

Lead Guest Editor, Electronics (MDPI) — SCIE Indexed Journal

Special Issue Title: Wireless Communications for Internet of Things and Vehicular Networks
Status: Special Issue Announced and Open for Submissions
Link: View Special Issue Webpage

- Serving as Guest Editor for a special issue focusing on IoT connectivity, vehicular networking, and intelligent wireless communication technologies.
- Overseeing manuscript screening, reviewer assignment, and quality assurance to ensure strong scientific and industrial relevance.
- Engaging international researchers and practitioners to promote cutting-edge research contributions in IoT-enabled vehicular systems.

Associate Editor, Engineering Reports (Wiley) — SCIE Indexed Journal

Role: Associate Editor
Link: View Editorial Board Page

- Handling submissions in wireless communications, intelligent networked systems, IoT, cyber-physical systems, and emerging engineering domains.
- Managing peer-review workflows, reviewer assignments, and rigorous technical evaluation to ensure publication quality.
- Supporting authors and reviewers through constructive feedback and transparent editorial decision processes.

Technical Program Committee (TPC) Member

Conferences: IEEE ICUFN, IEEE ICAIIC, IEEE ICC, IEEE GLOBECOM, IEEE VTC

- Serving on TPCs for flagship IEEE conferences in wireless communications, networking, and vehicular systems.
- Reviewing high-impact research papers related to V2X, 5G/6G networks, NTN integration, IoT, and AI-driven communication frameworks.
- Contributing to session organization, technical quality assurance, and research community engagement.

ACADEMIC BACKGROUND

- **Masters Combined Ph.D. in Computer Science and Engineering**
  Institution: Kyungpook National University, Daegu, South Korea
  Duration: 2019–2024 **CGPA:** 3.86/4.3
  Thesis Title: "Knowledge-Empowered Distributed Resource Allocation for NR-V2X Mode 2: Leveraging Deep Reinforcement Learning."
- **Bachelor of Science in Electrical Engineering**
  Institution: Institute of Space Technology, Islamabad, Pakistan
  Duration: 2014–2018 **CGPA:** 3.02/4.0
  Thesis Title: "Vehicle-to-Vehicle Communication Using Software-Defined Radios."

PATENTS

Context-Aware Resource Allocation Method and Supporting System
Patent No.: 10-2025-0007889 **Filed:** Jan 20, 2025

- Proposes a context-aware radio resource allocation technique for 5G/6G communication systems.
- Integrates environmental dynamics, mobility awareness, and traffic density information to enhance allocation decisions.
- Improves latency, reliability, and spectrum efficiency under highly dynamic connectivity conditions.
- Applicable to advanced 6G verticals, including cooperative mobility and intelligent wireless networks.

AI-Enabled Onboard Unit for Adaptive Resource Scheduling in NR-V2X Systems
Patent No.: 10-2025-0006853 **Filed:** Jan 16, 2025

- Introduces an AI-enabled Onboard Unit (OBU) architecture for adaptive resource scheduling in NR-based cellular systems.
- Utilizes LSTM prediction and deep reinforcement learning to estimate traffic demand and optimize radio resource usage in real time.
- Enables cooperative sensing, prioritizes mission-critical messages, and reduces energy consumption.
- Designed to support emerging 6G reliability and autonomy requirements.

Device and Method for Transmitting Messages between a Vehicle Terminal and a Wireless Terminal
Patent No.: 10-2024-0011319 **Filed:** Oct 2024

- Enables low-latency and high-accuracy message delivery by leveraging direct proximity-based communication rather than network-assisted links.
- Incorporates GNSS-based absolute positioning combined with location data shared over Wi-Fi Aware to calculate distance between vehicle and pedestrian terminals.
- Supports safety alerts for pedestrians and cooperative mobility by notifying road users when vehicles approach within a predefined distance threshold.

VOLUNTEERING ACTIVITIES

- Attended **Summer School for Research Collaboration** at the **University of Glasgow**, **Coventry University**, and **University of Strathclyde**, UK (August 2024).
- Represented **Kyungpook National University** in **3GPP Standardization Meetings**.
- Participated in **3GPP RAN Working Group 1, 2, and 4 (RAN-WG1 / RAN-WG2 / RAN-WG4)** discussions and technical sessions.

- Served as a **Reviewer** for reputed journals under **IEEE**, **ACM**, and **Elsevier**.
- Served as a **Technical Program Committee (TPC) Member** for IEEE and ACM flagship international conferences.

SELECTIVE PEER-REVIEWED PUBLICATIONS

1) Saad, Malik Muhammad, et al., *"Tiny Reptile Assisted Meta Learning with Attention Mechanism for Context-Aware Resource Allocation in NR-V2X,"* submitted to **IEEE Transactions on Intelligent Transportation Systems.**

2) Saad, Malik Muhammad, et al., *"Advancements in Vehicular Communications Technologies: C-V2X and NR-V2X Comparison,"* **IEEE Communications Magazine,** August 2021.

3) Saad, Malik Muhammad, et al., *"Non-Terrestrial Networks: An Overview of 3GPP Release 17 & 18,"* **IEEE Internet of Things Magazine,** Volume 7, Issue 1, January 2024.

4) Saad, Malik Muhammad, et al., *"Towards Efficient Mobility Management in 5G-Advanced: A Predictive Model for Network Slice Availability,"* **ACM SAC 2025.**

5) Saad, Malik Muhammad, et al., *"Federated Multi-Agent Reinforcement Learning for Resource Allocation in NR-V2X Mode 2,"* **IEEE Internet of Things Journal,** March 2025.

6) Saad, Malik Muhammad, et al., *"Twin Delayed DDPG (TD3)-based Edge Server Selection for 5G-Enabled Industrial and C-ITS Applications,"* **IEEE Open Journal of the Communications Society,** February 2025.

7) Saad, Malik Muhammad, et al., *"Knowledge-Empowered Distributed Learning Platform in Internet of Unmanned Aerial Agents to Support NR-V2X Communication,"* **IEEE Internet of Things Journal,** January 2025.

8) Saad, Malik Muhammad, et al., *"Analytical Models for Distributed Resource Allocation in NR-V2X Mode 2,"* **IEEE Internet of Things Journal,** 2023.

9) Saad, Malik Muhammad, et al., *"Age-of-Information Aware Intelligent MAC for Congestion Control in NR-V2X,"* **ICUFN 2023,** Paris, France, July 2023.

10) Saad, Malik Muhammad, et al., *"Air-Interface Virtualization Using Filter Bank Multicarrier and Orthogonal Frequency Division Multiplexing Configurations,"* **Transactions on Emerging Telecommunications Technologies,** 2020.

Note: A complete list of publications can be found on my Google Scholar profile:
View Google Scholar Profile

BOOK CHAPTERS

1) Malik Muhammad Saad, et al., *"AI-enabled Dynamic Resource Scheduling in Integrated TN-NTN Systems,"* in the book **"Artificial Intelligence for Integrated and Non-Terrestrial Networks,"** Springer Nature.

2) Malik Muhammad Saad, et al., *"Elevating Connectivity: Integrating Satellite Access Networks in 3GPP NTN Vision,"* in the book **"Integrated Terrestrial and Non-Terrestrial Networks,"** Springer Nature.

REFERENCES

Prof. Dongkyun Kim
Ph.D. from Seoul National University & Postdoc from University of California, USA
Position: Professor & Head, School of Computer Science and Engineering
Institution: Kyungpook National University, Daegu, South Korea

Email: dongkyun@knu.ac.kr
Relation to the Applicant: Ph.D. Academic Advisor
Contact: +82 10 9493 0001

Prof. Sobia Jangsher
Ph.D. from The University of Hong Kong (HKU), Hong Kong & Postdoc from Khalifa University, Abu Dhabi, UAE.
Position: Assistant Professor, School of Electronic Engineering
Institution: Dublin City University, Ireland
Email: sobia.jangsher@dcu.ie
Relation to the Applicant: Undergraduate Supervisor & Research Collaborator
Contact: +353 01 700 7666

Prof. Muhammad Ali Jamshed
Ph.D. from University of Surrey, United Kingdom
Position: Communication, Sensing and Imaging (CSI) School
Institution: University of Glasgow, United Kingdom
Email: MuhammadAli.Jamshed@glasgow.ac.uk
Relation to the Applicant: Research Collaborator
Contact: +44 7956 528927